

All That You Can Do *with* Google Analytics, and More

We have all heard about or used Google Analytics (GA) — the most popular tool to track user activity such as, but not limited to, page visits. Its utility and popularity means that everybody wishes to use it. This article focuses on how to use it correctly in a world where single page Angular and React applications are becoming more popular by the day.



A pertinent question is how does one track page visits in applications that have just one page?

As always, there are ways around this and we will look at one such option in this article. While we will do the implementation in an Angular application, the usage and concepts aren't very different if the app is in React. So, let's get started!

Getting the application ready

Getting a tracking ID: Before we write actual code, we need to get a tracking ID, the unique identifier that tells Google Analytics that data like a click or a page view is coming from a particular application.

To get this, we do the following:

1. Go to <https://analytics.google.com>.
2. Sign up by entering the required details. Make sure the registration is for the Web — ours is a Web application, after all.
3. Agree to the *Terms and*

Conditions, and generate your tracking ID.

4. Copy the ID, which will perhaps look something like 'UA-ID-Y'. Now that the ID is ready, let's write some code.

Adding *analytics.js* script

While the team at Google has done all the hard work to get the Google Analytics tools ready for us to use, this is where we do our part — make it available to the application. This is simple — all that's to be done is to add the following script to your application's *index.html*:

```
<script>
  (function(i,s,o,g,r,a,m){i['Google
AnalyticsObject']=r;i[r]=i[r]||function(){
    (i[r].q=i[r].q||[]).
push(arguments)},i[r].l=1*new
Date();a=s.createElement(o),
  m=s.getElementsByTagName(o)
[0];a.async=1;a.src=g;m.parentNode.
```

```
insertBefore(a,m)
})(window,document,'script','h
tps://www.google-analytics.com/
analytics.js','ga');
</script>
```

With that out of the way, let's see how we can initialise Google Analytics in the application.

Creating the tracker

Let's now set up the tracker for the application. For this, open *app.component.ts* and perform the following steps:

1. Declare a global variable named *ga*, of type *any* (remember, this is Typescript and you need to mention types).
2. Insert the following line of code into *ngOnInit()* of your component:

```
ga('create', YOUR_TRACKING_ID, 'auto');
```

Congratulations! You have now successfully initiated a Google